

Toward Pervasive Skin Health: Cross-Domain Learning for Segmentation of Mobile-Acquired Inflammatory Diseases

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Abstract

Recent advances in deep learning have yielded impressive skin-lesion segmentation performance, yet progress has centred almost exclusively on dermoscopic images of pigmented skin lesions. Inflammatory conditions such as urticaria and psoriasis remain under-represented especially when lesions are photographed with consumer smartphones, where colour casts, motion blur and framing variability further erode the already limited training data. In this extended-abstract we position our work as the first to demonstrate that cross-disease transfer learning and domain-adversarial adaptation from large dermoscopic datasets (ISIC, HAM10000) can substantially boost pixel-level segmentation accuracy on two extremely small mobile-acquired datasets (urticaria $n = 150$, psoriasis $n = 500$). By narrowing the performance gap, our approach lays the groundwork for these automated tools to effectively monitor chronic skin disorders ‘in the wild,’ leveraging ubiquitous mobile technology to extend their reach into patients’ daily lives and diverse real-world environments.

CCS Concepts

• **Human-centered computing** → **HCI applications**; • **Applied computing** → **Health informatics**; • **Computing methodologies** → **Image segmentation**; *Transfer learning*.

Keywords

skin lesion segmentation, deep learning, transfer learning, domain adaptation, mobile health, teledermatology, inflammatory skin diseases, psoriasis, urticaria, computer vision, medical imaging

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1 Introduction

Inflammatory dermatoses such as chronic urticaria and psoriasis impose a psychosocial burden rivalling many systemic diseases and affect a substantial share of the over 3 billion people living with skin conditions worldwide[1]. Reliable, fine-grained lesion segmentation is essential for downstream tasks such as objective surface-area estimation, treatment monitoring, and teledermatology triage.

However, despite the growing popularity of deep learning-based segmentation, most high-performance models are trained on dermoscopic images of pigmented lesions (e.g., ISIC[5], HAM10000[21]) and rely on thousands of densely annotated samples. In contrast, segmentation research on inflammatory diseases—especially those captured via consumer smartphones—is significantly underexplored due to a lack of large, labeled datasets. This gap is further exacerbated by privacy concerns, annotation costs, and variability in image quality and framing. Addressing this challenge is critical for enabling real-world applications in mobile teledermatology and expanding equitable access to dermatologic care.

We address this imbalance by proposing a **three-stage cross-domain learning pipeline** to boost segmentation accuracy on mobile-acquired inflammatory diseases. Our approach first leverages self-supervised representation learning with BYOL[11] on a large corpus of unlabeled dermoscopic and clinical images to initialize a disease-agnostic encoder. We then perform supervised fine-tuning on the HAM10000 skin cancer dataset, followed by domain-adversarial training using CDAN[16] to adapt to small, smartphone-acquired datasets of urticaria and psoriasis. We specifically focus on chronic urticaria and psoriasis as they are highly prevalent inflammatory conditions that impose a significant patient burden, yet represent distinct visual challenges for automated analysis.

Our method demonstrates consistent improvements in segmentation performance, particularly in low-data regimes, and generalizes effectively across disease types. These findings suggest that our framework can serve as a practical blueprint for clinicians and researchers aiming to develop robust lesion segmentation models under limited annotation budgets and real-world imaging constraints.

2 Related Work

2.1 Segmentation in Dermatology

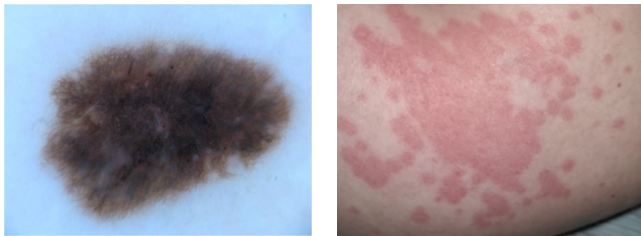
Fully convolutional networks like U-Net and its variants [20, 23] form the backbone of competitive segmentation systems for dermoscopic images, while inflammatory dermatoses receive far less attention. For these conditions, pixel-level masks are rarely available, and data scarcity, particularly for smartphone-captured images, is a major bottleneck. For instance, efforts to develop accurate segmentation models for urticaria [15, 17] and psoriasis [7, 19] have been hampered by the lack of large, public datasets.

2.2 Self-Supervised Learning for Medical Segmentation

Self-supervised learning (SSL) is a method to enhance downstream task capabilities, such as segmentation, without requiring annotated masks, making it ideal for the data-scarce medical field. Studies have shown that SSL can boost segmentation accuracy for skin lesions [4]. However, existing literature has mainly focused on dermoscopic pigmented lesion datasets. This work is among the first to explore the effectiveness of SSL in smartphone-acquired datasets of inflammatory dermatoses for segmentation.

2.3 Transfer Learning and Domain Adaptation for Medical Segmentation

Transfer Learning is another popular method utilized to boost segmentation accuracy in data-scarce scenarios. Karri et al. found efficacy in cross-domain transfer learning for segmenting pigmented lesion datasets (HAM10000 and MoleMap) [13]. However, it is a known challenge in medical imaging that transferring knowledge between distant domains—such as from dermoscopic images of pigmented lesions to mobile-acquired images of inflammatory lesions—can lead to performance degradation [18]. These domains differ drastically not only in imaging modality and quality (e.g., lighting, focus) but also in fundamental lesion characteristics (e.g., color and texture), as illustrated in Figure 1.



(a) Dermoscopic image of pigmented lesion from the ISIC 2019 Challenge dataset [5, 6, 21]. More at <https://challenge.isic-archive.com/data/>.
(b) Smartphone image of urticaria from DermNet NZ [9]. More at <https://dermnetnz.org/images>.

Figure 1: Comparison of dermatological images: (a) a dermoscopic image of a skin lesion and (b) a smartphone photograph of urticaria.

To address these limitations, **unsupervised domain adaptation (UDA)** offers an alternative to costly relabelling when models face cross-device or cross-site shift. Early gradient-reversal

networks (DANN) reduce global feature discrepancy, and more refined techniques such as Conditional Domain Adversarial Networks (CDAN) align category-specific distributions [10, 16]. While UDA has demonstrated effectiveness in skin lesion classification [3], its application to skin lesion segmentation, particularly for inflammatory conditions captured by smartphones, remains largely unexplored.

3 Methodology

Our aim is to improve segmentation performance on small smartphone-acquired datasets by leveraging public skin lesion datasets that diverge from the target domain. We first conduct self-supervised learning on unlabeled dermoscopic and clinical images to initialize a disease-agnostic encoder. We then perform supervised fine-tuning on the HAM10000 skin cancer dataset to leverage abundant mask annotation to further tune the model for skin lesion segmentation. Finally, we conduct domain-adversarial training to adapt to small, smartphone-acquired datasets. U-Net++ was chosen as our model backbone due to its capability over small datasets in the medical field. We employ 5 public datasets, summarized in table 1.

3.1 Stage 1 – Self-supervised representation learning

We first initialise a disease-agnostic encoder with **Bootstrap Your Own Latent (BYOL)** [11]. We embed an ImageNet-pretrained[8] ResNet-50 [12] inside a U-Net++ encoder to preserve the high-resolution feature maps required by pixel-wise segmentation. We conducted training on a joint dataset of DermNet and ISIC 2019 to sample pigmented and inflammatory lesions, as well as images from the smartphone and dermoscope domain.

3.2 Stage 2 – Supervised fine-tuning on HAM10000

We attach a lightweight 1×1 convolutional segmentation head to the pretrained encoder and fine-tune on the HAM10000 dermoscopic melanoma dataset. This allows the model to learn segmentation patterns for skin lesions by taking advantage of abundant lesion annotations. The checkpoint with the highest validation performance is used to initialize Stage 3.

3.3 Stage 3 – Domain-adversarial adaptation to smartphone imagery

To bridge the appearance gap between dermoscopes and phone cameras we adopt **Conditional Domain Adversarial Networks (CDAN)** [16].

Given encoder features $f \in \mathbb{R}^{C \times H \times W}$ and the soft mask $p \in [0, 1]^{1 \times H \times W}$, CDAN operates on the outer product $g = \text{vec}(f) \otimes \text{vec}(p)$, which captures class-conditional structure. A gradient-reversal layer (GRL) with coefficient λ_{GRL} feeds g to a domain discriminator D ; the overall loss

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \underbrace{\gamma \text{BCE}(D(g), y_d)}_{\mathcal{L}_{\text{dom}}}$$

Figure 2: Overview of our three-stage training pipeline.

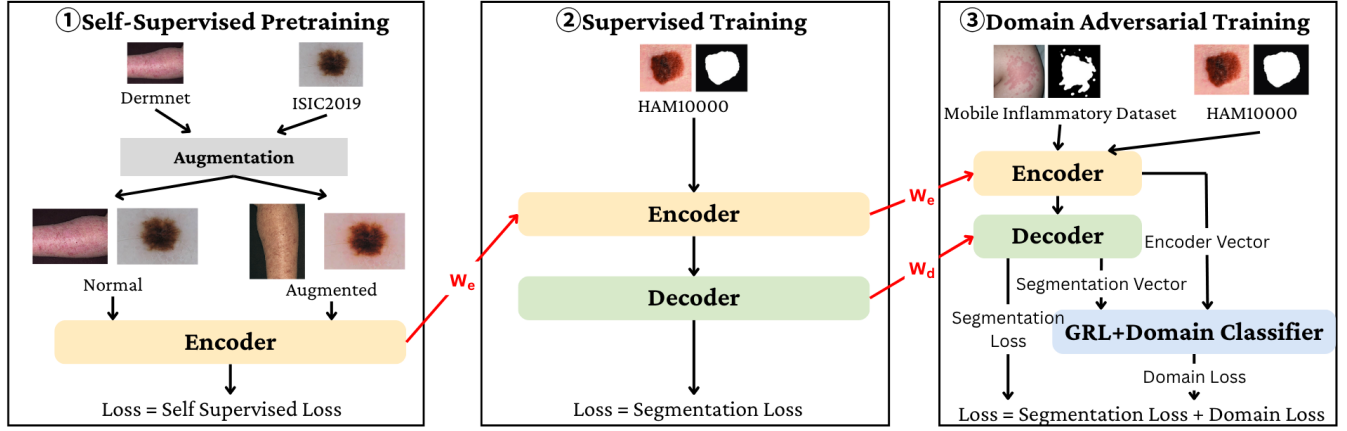


Table 1: Overview of datasets in each training stage.

Dataset	Stage	#Images	Annotation	Disease	Modality
DermNet	Stage 1	19,500	—	Pigmented and inflammatory lesions	Smartphone
ISIC 2019	Stage 1	25,000	—	Pigmented lesions	Dermoscope
HAM10000	Stage 2 and 3	10,000	Pixel-wise lesion masks	Pigmented lesions	Dermoscope
Roboflow Urticaria	Stage 3	150	Pixel-wise lesion masks	Inflammatory lesions	Smartphone
Roboflow Psoriasis	Stage 3	500	Pixel-wise lesion masks	Inflammatory lesions	Smartphone

aligns the feature distributions while preserving segmentation quality.

We also employ architecture refinements such as **Domain-specific GroupNorm** which replaces batch-norm in the bottleneck; source and target statistics are accumulated separately to avoid leakage. Furthermore, A **CBAM** attention block[22] enhances salient inflammatory patterns before pooling.

4 Evaluation

The evaluation is performed on two smartphone-acquired inflammatory disease datasets: The Roboflow Urticaria dataset [14] and the Roboflow Psoriasis dataset [2]. A 5-fold cross-validation scheme with 80-20 split is employed for Stage 3 on the target datasets. The checkpoint with the highest validation F1-score (Dice coefficient) within each fold is selected for reporting.

4.1 Experimental Comparisons

- (1) **Proposed Method vs. Baseline:** The core comparison is against a baseline U-Net++ model (with an ImageNet-pretrained encoder) that is directly fine-tuned on the target dataset (Urticaria or Psoriasis) without the proposed Stage 1, 2, or 3 enhancements. Furthermore, we repeat the experiment using different fractions of the target training data to assess data efficiency.
- (2) **Ablation Studies:** The contribution of individual components of the pipeline is assessed by systematically removing these stages in separate experimental runs.

4.2 Evaluation Metrics

Performance is quantified using standard segmentation metrics, computed on the validation sets:

F1-score (Dice): Measures the overlap between predicted and ground truth masks, calculated as:

$$F1 = \frac{2 \times |\text{Predicted} \cap \text{Ground Truth}|}{|\text{Predicted}| + |\text{Ground Truth}|} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

where TP, FP, and FN are true positives, false positives, and false negatives at the pixel level, respectively.

Intersection over Union (IoU) / Jaccard Index: Also measures overlap, defined as:

$$IoU = \frac{|\text{Predicted} \cap \text{Ground Truth}|}{|\text{Predicted} \cup \text{Ground Truth}|} = \frac{TP}{TP + FP + FN}$$

The mean and standard deviation of these metrics across the 5 folds are reported.

5 Results

5.1 Comparative Performance on Target Datasets

Our proposed method demonstrates consistent and significant improvements over the baseline across both datasets and varying training data sizes. The advantages are particularly pronounced in low-data regimes, showcasing the data efficiency of our approach.

On the Psoriasis dataset (Table 2), the proposed method consistently achieved higher IoU and F1-scores. The benefits become even more stark with limited data: using only 40 Psoriasis images, our

method achieved absolute IoU gain of 13.14% and absolute F1-score gain of 9.56%.

Table 2: Comparison of segmentation performance (Mean $\pm$ Std Dev % over 5 folds) on the Roboflow Psoriasis dataset

Training Size	IoU (%)		F1-score (Dice) (%)	
	Baseline	Proposed	Baseline	Proposed
400	77.68 $\pm$ 1.51	81.09 $\pm$ 1.20	87.37 $\pm$ 0.94	89.47 $\pm$ 0.74
300	75.31 $\pm$ 2.96	80.50 $\pm$ 1.27	85.83 $\pm$ 1.93	89.12 $\pm$ 0.78
200	72.95 $\pm$ 2.81	79.89 $\pm$ 0.81	84.27 $\pm$ 1.89	88.73 $\pm$ 0.47
120	71.22 $\pm$ 4.41	78.70 $\pm$ 1.19	83.02 $\pm$ 3.17	87.98 $\pm$ 0.75
80	72.70 $\pm$ 1.44	75.94 $\pm$ 1.48	84.11 $\pm$ 0.97	86.20 $\pm$ 0.95
40	60.93 $\pm$ 7.02	74.07 $\pm$ 2.52	75.38 $\pm$ 5.49	84.94 $\pm$ 1.70

Similar substantial gains were observed on the Urticaria dataset (Table 3), which is inherently more challenging, as reflected by the lower overall scores for both methods. The visual characteristics of urticaria—such as its transient nature, less defined borders, and greater variability in lesion shape and erythema compared to the more distinct plaques of psoriasis—present a greater challenge for segmentation algorithms. Our method achieved a significant performance boost, with its effectiveness further emphasized in the 40-image scenario. Here, our method increased the absolute IoU by 12.35% and absolute F1-score by 13.03%. These results highlight the robustness and adaptability of our multi-stage pipeline.

Table 3: Comparison of segmentation performance on the Roboflow Urticaria dataset

Training Size	IoU (%)		F1-score (Dice) (%)	
	Baseline	Proposed	Baseline	Proposed
120	42.09 $\pm$ 2.94	51.34 $\pm$ 1.58	59.11 $\pm$ 2.95	67.63 $\pm$ 1.27
80	40.09 $\pm$ 3.28	50.07 $\pm$ 1.13	57.01 $\pm$ 3.40	66.54 $\pm$ 1.27
40	30.58 $\pm$ 3.18	42.93 $\pm$ 4.19	46.65 $\pm$ 3.57	59.68 $\pm$ 3.89

5.2 Ablation Studies

Table 4: Ablation study results

Method Configuration	Psoriasis (400)		Psoriasis (120)		Urticaria (120)	
	IoU (%)	F1-score (%)	IoU (%)	F1-score (%)	IoU (%)	F1-score (%)
Baseline (ImageNet FT)	77.68 $\pm$ 1.51	87.37 $\pm$ 0.94	71.22 $\pm$ 4.41	83.02 $\pm$ 3.17	42.09 $\pm$ 2.94	59.11 $\pm$ 2.95
+BYOL (S1) only + FT	80.97 $\pm$ 1.47	89.44 $\pm$ 0.89	75.31 $\pm$ 2.46	85.85 $\pm$ 1.62	36.43 $\pm$ 1.34	53.13 $\pm$ 2.24
+HAM (S2) only + FT	79.76 $\pm$ 1.32	88.43 $\pm$ 0.84	73.69 $\pm$ 1.95	84.78 $\pm$ 1.30	30.71 $\pm$ 1.18	46.76 $\pm$ 2.14
+CDAN (S3) only + FT	80.13 $\pm$ 1.37	88.90 $\pm$ 0.84	75.13 $\pm$ 1.14	85.69 $\pm$ 0.78	43.13 $\pm$ 0.98	59.51 $\pm$ 1.94
Full Proposed	81.09 $\pm$ 1.20	89.47 $\pm$ 0.74	78.70 $\pm$ 1.19	87.98 $\pm$ 0.75	51.34 $\pm$ 1.58	67.63 $\pm$ 1.27

To dissect the contribution of each component in our pipeline, we conducted ablation studies. On the Psoriasis dataset, each individual stage of our methodology generally demonstrated a positive contribution to performance. While these individual stages showed somewhat comparable efficacy, the full proposed pipeline (S1+S2+S3) consistently achieved the highest scores. This indicates a synergistic effect where combining all stages yields the optimal outcome for Psoriasis segmentation.

The ablation results for the Urticaria dataset with 120 training images revealed different dynamics, likely due to a stronger

domain shift and the inherent difficulty of the dataset. Interestingly, applying only Stage 1 or only Stage 2 led to a performance degradation compared to the baseline. This suggests that the representations learned from DermNet/ISIC (S1) or HAM10000 (S2) were not directly beneficial and possibly detrimental for Urticaria when not integrated into the full adaptation pipeline. Critically, the full proposed method (S1+S2+S3) significantly outperformed all other configurations, achieving an IoU of 51.34%. This highlights that the complete, sequential multi-stage approach is essential for robust performance on more challenging domains like Urticaria.

5.3 Training Time Evaluation

The training times on an NVIDIA A100 GPU are detailed in Table ???. While our Stage 3 adaptation adds a modest overhead compared to baseline fine-tuning (e.g., 37 vs. 5 minutes for Psoriasis), the lengthy initial pre-training (Stages 1 and 2) is a one-time cost, as the resulting weights can be shared. Critically, inference time is unaffected because the model architecture remains unchanged, a key factor for practical deployment.

Table 5: Comparison of training times on NVIDIA A100 GPU.

Task	Method / Stage	Duration
<i>Common Pre-training</i>		
Stage 1	Self-Supervision	6h 43m
Stage 2	Supervised Fine-tuning	1h 27m
<i>Target Dataset Adaptation</i>		
Urticaria	Baseline Fine-tuning	2m
	Proposed Stage 3	13m
Psoriasis	Baseline Fine-tuning	5m
	Proposed Stage 3	37m

6 Conclusion

This work presents a three-stage cross-domain learning pipeline that significantly enhances the segmentation of inflammatory dermatoses (psoriasis and urticaria) from smartphone images, a crucial task hampered by the scarcity of labeled mobile-acquired data.

Although our framework is designed for small target datasets, its robustness needs to be confirmed with more diverse, "in-the-wild" clinical data. Future work will prioritize extensive real-world testing using prospectively collected clinical data from diverse patient populations, varying smartphone devices and diseases.

Overall, our multi-stage cross-domain learning pipeline offers a practical and data-efficient solution for a pressing challenge in mobile tele dermatology. Such tools have the potential to improve objective disease assessment and treatment monitoring for inflammatory skin conditions, thereby broadening access to specialized dermatological insights, particularly in settings with limited resources or access to specialists.

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